Math 3A Problem Set 8 Integration Practice Name Wes Oakley

For u-substitution, be sure to write down what u and du are equal to.

1) $\int \frac{2x}{\sqrt{9-x^2}}\,dx$

$u = 9 - x^2$

$\frac{d}{dx} u = 9 - x^2 \frac{d}{dx}$

$\frac{du}{dx} = -2x$

$du = -2x\,dx$

$dx = \frac{du}{-2x}$

$\int \frac{2x}{\sqrt{u}}\,dx$

$\int 2x\, u^{-1/2}\,dx$

$\int 2x\, u^{-1/2} \left(\frac{du}{-2x}\right)$

$\int \cancel{2x}\, u^{-1/2} (-1)\left(\frac{1}{\cancel{2x}}\right) du$

$-\int u^{-1/2}\,du$

$-\frac{u^{1/2}}{1/2} + C = -2u^{1/2} + C$

$= \boxed{-2\sqrt{9-x^2} + C}$

2) $\int \frac{3x^4 - 9x^2 + 1}{x^2}\,dx$

$\int \frac{3x^4}{x^2} - \frac{9x^2}{x^2} + \frac{1}{x^2}\,dx$

$\int 3x^2 - 9 + x^{-2}\,dx$

$= 3\frac{x^3}{3} - 9x + \frac{x^{-1}}{-1} + C$

$= x^3 - 9x - x^{-1} + C$

$= \boxed{x^3 - 9x - \frac{1}{x} + C}$

3) $\int \sin x \cos x\,dx$

Since cos is the derivative of sin

$\int u\,u'\,dx \longrightarrow \int u\,du$ (let $u = \sin x$)

$\frac{d}{dx} u = \frac{d}{dx} \sin x$

$\frac{du}{dx} = \cos x$

$du = \cos x\,dx$

$\int \sin x \cos x\,dx \Rightarrow \int u\,du$

$= \frac{u^2}{2} + C$

$\boxed{= \frac{\sin^2 x}{2} + C}$

4) $\int \sin^2 x \cos x\,dx$

$u = \sin x$

$\frac{d}{dx} u = \sin x \frac{d}{dx}$

$\frac{du}{dx} = \cos x$

$du = \cos x\,dx$

$dx = \frac{du}{\cos x}$

$\int u^2 \cancel{\cos x} \frac{du}{\cancel{\cos x}}$

$\int u^2\,du$

$= \frac{u^3}{3} + C$

$= \boxed{\frac{1}{3}\sin^3 x + C}$

5) $\int \frac{\sin x}{\sqrt{\cos x}}\,dx$

$u = \cos x$

$\frac{d}{dx} u = \cos x \frac{d}{dx}$

$\frac{du}{dx} = -\sin x$

$du = -\sin x\,dx$

$dx = \frac{du}{-\sin x}$

$\int \frac{\sin x}{\sqrt{u}} \cdot \left(\frac{du}{-\sin x}\right)$

$\int \cancel{\sin x}\, u^{-1/2} (-1)\left(\frac{1}{\cancel{\sin x}}\right) du$

$-\int u^{-1/2}\,du$

$= -\frac{u^{1/2}}{1/2} + C$

$= -2u^{1/2} + C$

$= \boxed{-2\sqrt{\cos x} + C}$

6) $\int \frac{\sin x}{1+\cos^2 x}\,dx$

$u = \cos x$

$\frac{d}{dx} u = \cos x \frac{d}{dx}$

$\frac{du}{dx} = -\sin x$

$du = -\sin x\,dx$

$dx = \frac{du}{-\sin x}$

$\int \frac{\cancel{\sin x}}{1 + u^2} \cdot (-1)\left(\frac{1}{\cancel{\sin x}}\right) du$

$-\int \frac{1}{1+u^2}\,du$

$= -\tan^{-1}(u) + C$

$= \boxed{-\tan^{-1}(\cos x) + C}$

7) Integrate to find the area between $g(x) = \sin(x)$ and $f(x) = cos(x)$ on the interval $\left[0, \frac{\pi}{4}\right]$.

$y = \sin x$, $y = \cos x$

on this interval cos is above sin

$$\text{Area} = \int_a^b (\text{top} - \text{bottom})\, dx$$

$$\text{Area} = \int_0^{\pi/4} (\cos x - \sin x)\, dx$$

$$= \sin x + \cos x \Big|_0^{\pi/4}$$

Upper bound: $\sin\left(\frac{\pi}{4}\right) + \cos\left(\frac{\pi}{4}\right) = \frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2} = \sqrt{2}$

Lower bound: $\sin(0) + \cos(0) = 0 + 1 = 1$

$$\boxed{\text{Area} = \sqrt{2} - 1}$$

8) Integrate to find the area above the x-axis and below $f(x) = -x^2 + 5x - 4$.

I need to know the x intercepts:

$$-x^2 + 5x - 4 = 0$$

$$-(x-1)(x-4) = 0$$

$x = 1$ $\quad$ $x = 4$

$$\text{Area} = \int_1^4 (-x^2 + 5x - 4)\, dx$$

$$= -\frac{x^3}{3} + \frac{5x^2}{2} - 4x \Big|_1^4$$

Upper bound − Lower bound

$$-\frac{4^3}{3} + \frac{5(4)^2}{2} - 4(4) \qquad -\frac{1^3}{3} + \frac{5(1)^2}{2} - 4(1)$$

$$-\frac{64}{3} + 40 - 16 \qquad -\frac{1}{3} + \frac{5}{2} - 4$$

$$\frac{8}{3} \qquad -\frac{11}{6}$$

$$\text{Area} = \frac{8}{3} - \left(-\frac{11}{6}\right)$$

$$= \frac{16}{6} + \frac{11}{6} = \frac{27}{6} = \boxed{\frac{9}{2}}$$

9) Integrate to find the finite area between $f(x) = \frac{1}{x}$ and $g(x) = 5 - 2x$.

$y = \frac{1}{x}$, $y = 5 - 2x$

we need to find where they intersect

$$\frac{1}{x} = 5 - 2x$$

$$1 = 5x - 2x^2$$

$$2x^2 - 5x + 1 = 0$$

$$x = \frac{5 \pm \sqrt{25 - 8}}{4}$$

$$x = \frac{5 \pm \sqrt{17}}{4}$$

So the bounds are

$a = \frac{5 + \sqrt{17}}{4}$, $b = \frac{5 - \sqrt{17}}{4}$

$$\text{Area} = \int_a^b 5 - 2x - \frac{1}{x}\, dx$$

Integrate

$$\int 5 - 2x - \frac{1}{x}\, dx$$

$$f(x) = 5x - x^2 - \ln|x|$$

$$\text{Area} = f(b) - f(a)$$

This is difficult for me to calculate by hand

I used a calculator which gave me 2.811

Input:	Decimal approximation:
$\left(5\left(\frac{1}{4}(5+\sqrt{17})\right) - \left(\frac{1}{4}(5+\sqrt{17})\right)^2 - \log\left(\left\lvert\frac{1}{4}(5+\sqrt{17})\right\rvert\right)\right) - \left(5\left(\frac{1}{4}(5-\sqrt{17})\right) - \left(\frac{1}{4}(5-\sqrt{17})\right)^2 - \log\left(\left\lvert\frac{1}{4}(5-\sqrt{17})\right\rvert\right)\right)$	2.81170302321371094

$$\text{Area} = 5x - x^2 - \ln|x| \Big|_{\frac{5-\sqrt{17}}{4}}^{\frac{5+\sqrt{17}}{4}}$$

upper bound:

$$f(b) = 5\left(\frac{5+\sqrt{17}}{4}\right) - \left(\frac{5+\sqrt{17}}{4}\right)^2 - \ln\left|\left(\frac{5+\sqrt{17}}{4}\right)\right|$$

$$\frac{25 + 5\sqrt{17}}{4} - \frac{(5+\sqrt{17})^2}{16}$$

$$\frac{100 + 20\sqrt{17}}{16} - \frac{42 + 10\sqrt{17}}{16}$$

$$\frac{58 + 10\sqrt{17}}{16}$$

$$\frac{29 + 5\sqrt{17}}{8} - \ln\left|\frac{5+\sqrt{17}}{4}\right|$$

lower bound

$$f(a) = 5\left(\frac{5-\sqrt{17}}{4}\right) - \left(\frac{5-\sqrt{17}}{4}\right)^2 - \ln\left|\left(\frac{5-\sqrt{17}}{4}\right)\right|$$